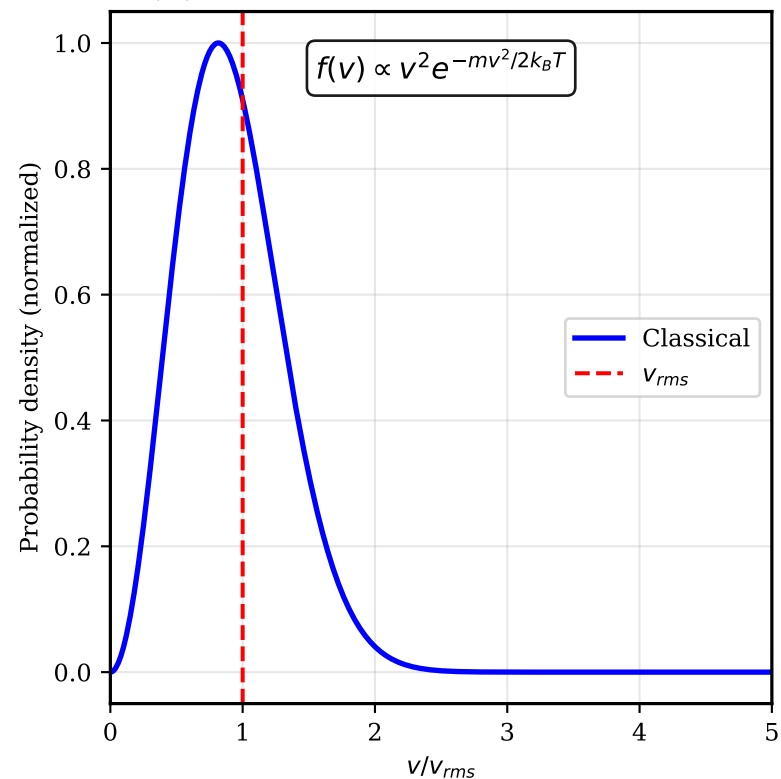
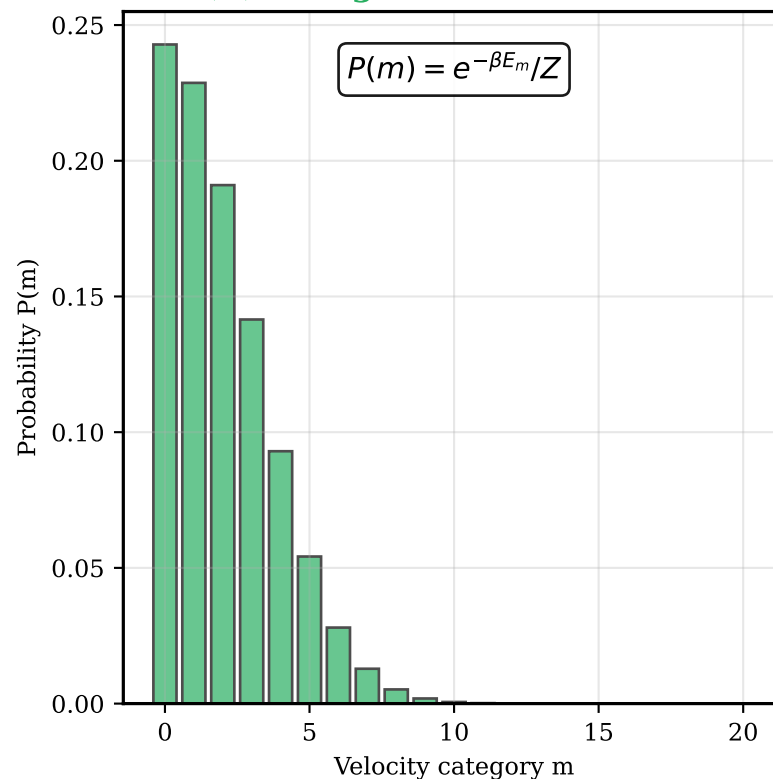
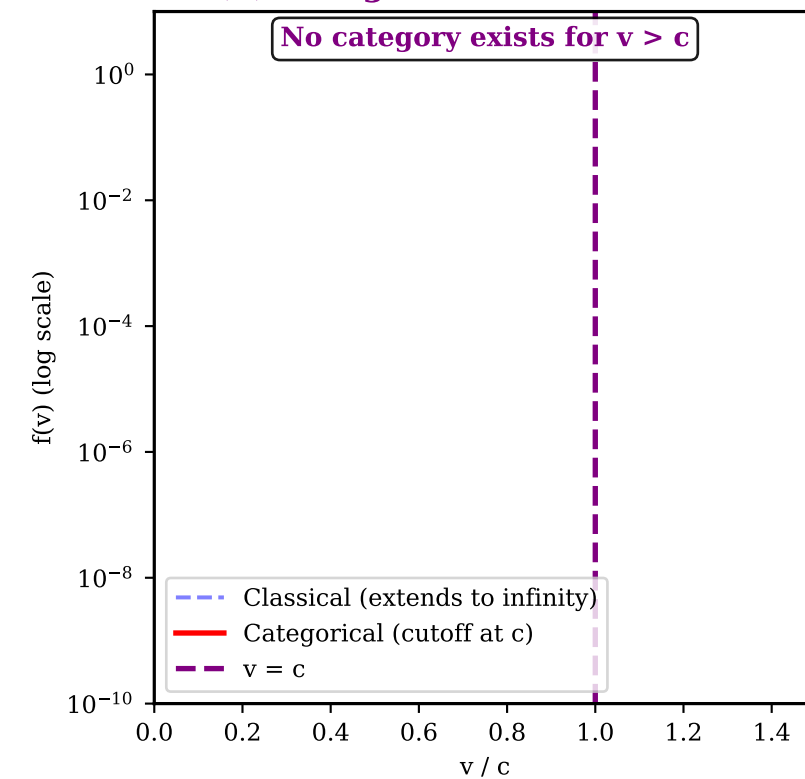
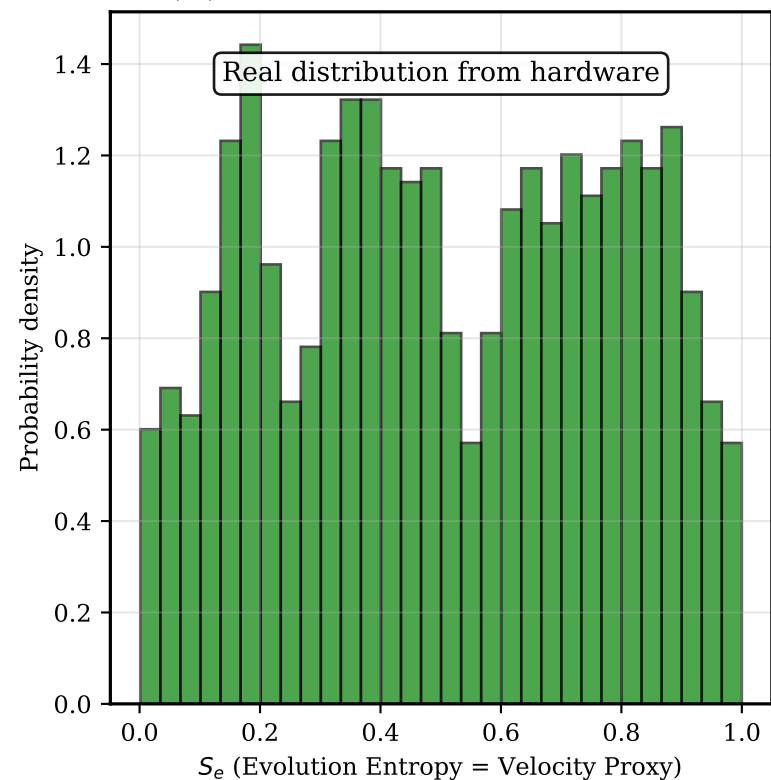
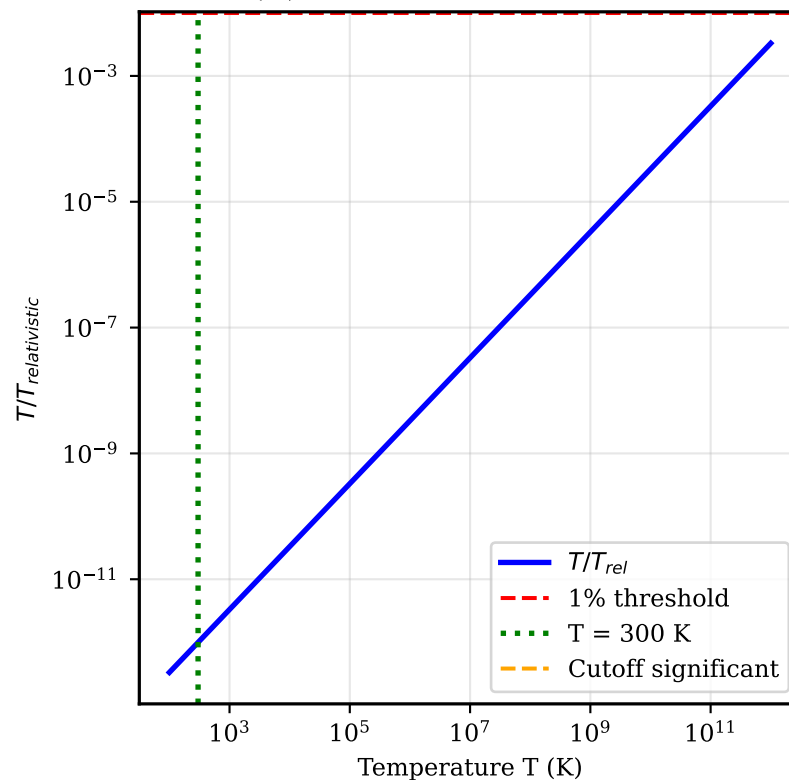


(A) Classical Maxwell-Boltzmann**(B) Categorical Distribution****(C) Categorical Cutoff at $v = c$** **(D) Hardware Gas Distribution****(E) When Cutoff Matters****(F) Maxwell-Boltzmann Summary**

MAXWELL-BOLTZMANN WITH CUTOFF
=====

Classical: $f(v) \sim v^2 \exp(-mv^2/2kT)$

- Extends to $v \rightarrow \infty$
- Violates relativity

Categorical: $P(m) = \exp(-\beta E_m) / Z$

- $m \in [0, 1, \dots, M_{\max}]$
- $M_{\max}$ corresponds to $v = c$
- Distribution intrinsically bounded

Key Insight:

- No category exists for $v > c$
- $P(v > c) = 0$ identically

At $T = 300$ K:

- $T/T_{\text{rel}} = 9.91e-13$
- Cutoff effect: NEGLIGIBLE

At $T \sim 10^{10}$ K:

- Cutoff becomes significant
- Must use categorical form

VALIDATION: PASS

The categorical framework automatically enforces relativistic causality.